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1. The 1 in 60 Rule — Concept & Application

✈ The Core Idea

For any angle up to about **20°**, if the adjacent side of a right-angled triangle is **60 units long**, then the length of the opposite side (in the same units) is *numerically equal* to the angle in degrees.

In flight: if you have travelled **60 NM** along-track and you are **N NM** off-track, your track error angle is approximately **N degrees**.

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Figure 10.1

Now raise a perpendicular at one end, exactly 10 mm high, and join them with a hypotenuse:

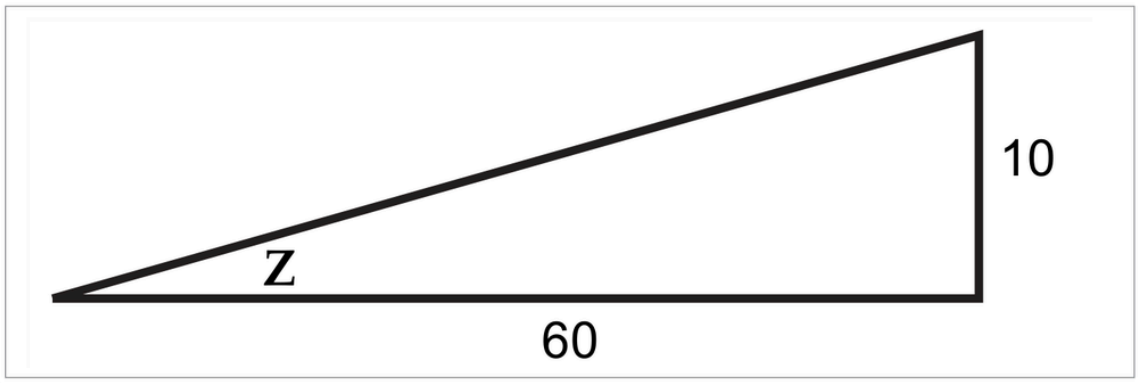


Figure 1 — The 1 in 60 Rule — a right-angled triangle with an adjacent side of 60 units: the angle z° equals the opposite side in the same units

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This is an enormously useful tool for in-flight calculations, removing the need for a protractor. The rule works because:

- When the adjacent = 60, tan(z°) ≈ z/60 for small angles (up to ~20°)
- Equivalently: one *radian* = 57.3° ≈ 60° for this approximation

2. Measuring Angles in Flight

Suppose you are flying on a planned track of **100°(T)**. After **60 NM** from your last on-track fix, you identify a ground feature that is **4 NM right of track**.

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> ✎ **Applying the rule directly:**
>
> Adjacent = 60 NM, Opposite = 4 NM → **Track Error Angle = 4°** right
>
> Your Track Made Good (TMG) = 100° + 4° = **104°(T)**

No protractor needed — simply read off the angle from the numbers. The next chapter covers what to do once you have the error angle.

3. Geometry of the 1 in 60 Rule — The Radian Derivation

Why does this rule work? The key lies in the definition of a **radian**.

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Imagine a circle of 1 metre radius.

What is the formula to calculate the circumference?

Circumference = $2 \pi r$

Taking π as 3.142, which is close enough for our purposes, then the

circumference = 2 × 3.142 × 1 = 6.284 metres.

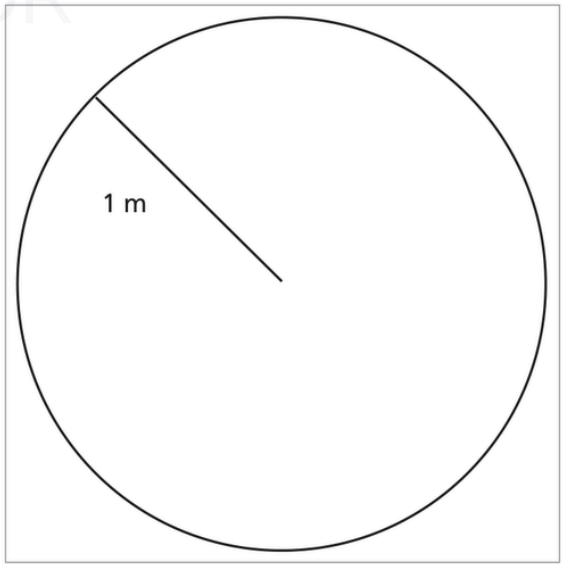


Figure 2 — A circle of 1 metre radius — travelling 1 metre along the circumference subtends angle θ at the centre

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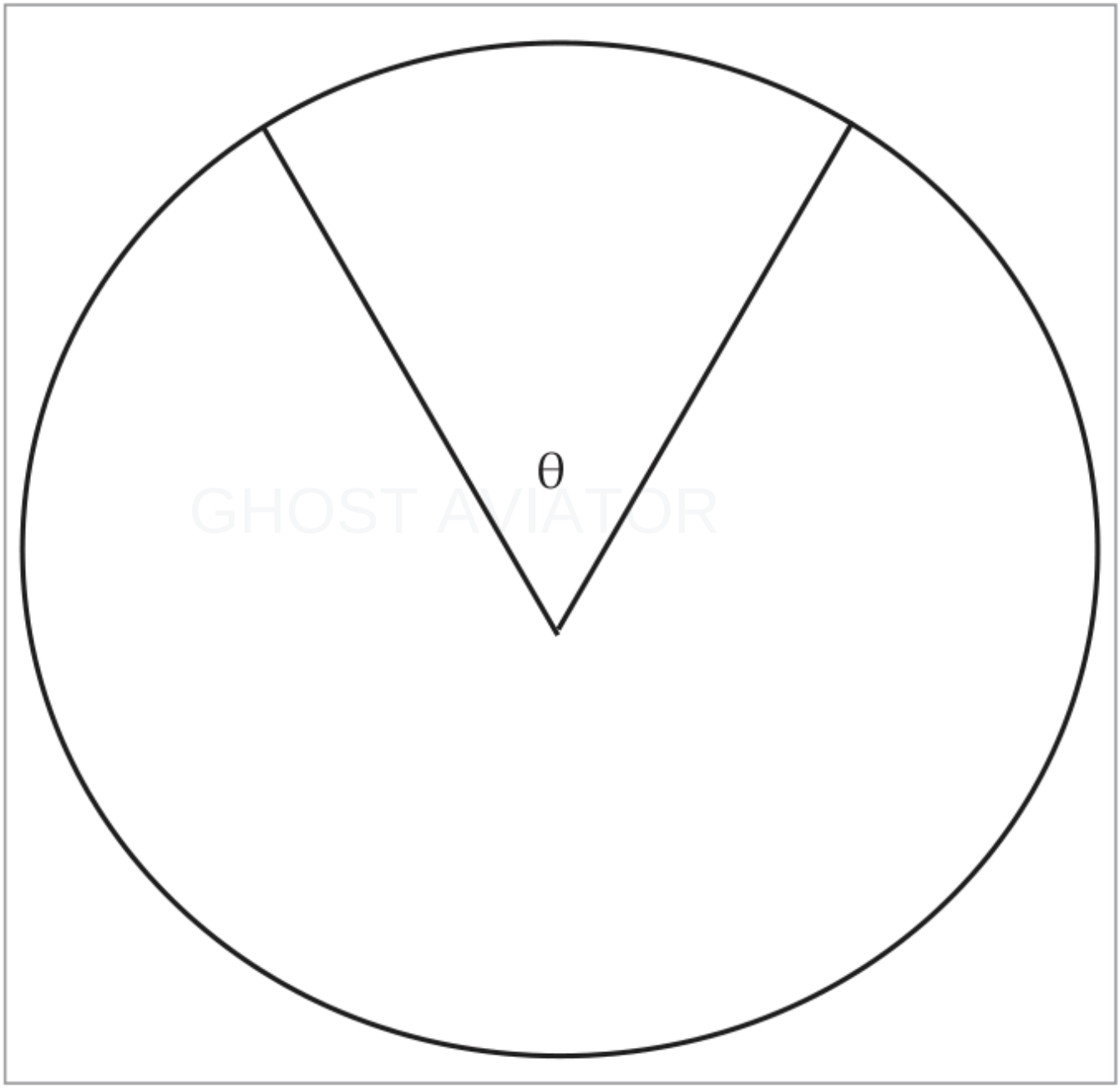


Figure 3 — The arc of 1 metre on a unit circle subtends θ = 57.3° at the centre (1 radian)

The Radian Calculation

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Step-by-step derivation

- **Circle circumference:** $2\pi r = 2 \times 3.142 \times 1 =$ **6.284 m**
- **Full circle = 360°** corresponds to 6.284 m of arc
- **1 metre of arc** corresponds to θ, where: $\theta/1 = 360°/6.284 \rightarrow$ **θ = 57.3°**
- **Scaling:** This ratio holds for any circle radius — 57.3° always corresponds to an arc = radius
- **Rounding:** 57.3 → **60** (introduces ~5% error — acceptable for flight calculations)

📚 Why 60 and not 57.3?

Strictly, this should be the "1 in 57.3 rule". But 57.3 is awkward for mental arithmetic at high workload in the cockpit. Rounding to **60** introduces only ~5% error, which is perfectly acceptable for navigation.

4. The Tangent Explanation

An alternative derivation shows why the rule works for a flat right-angled triangle (rather than an arc).

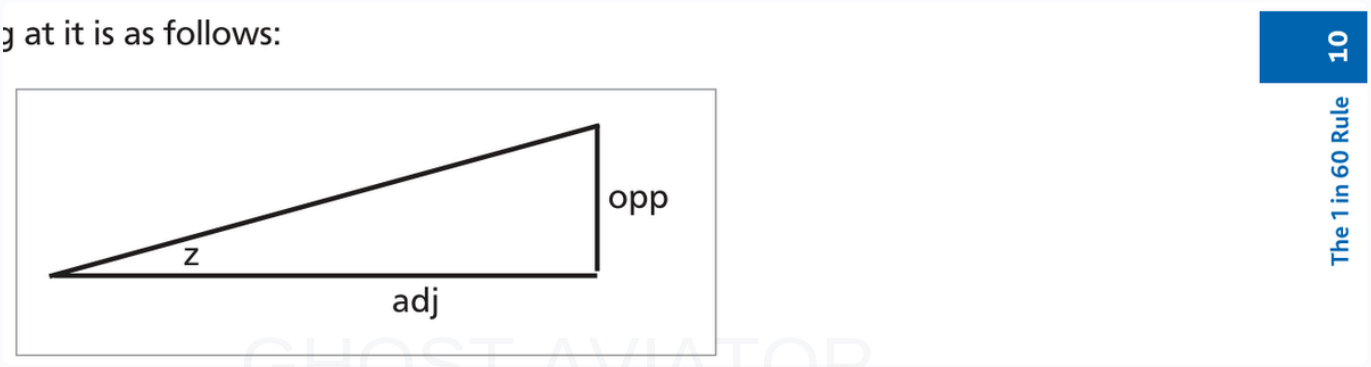


Figure 4 — Right-angled triangle with angle z — adjacent, opposite, and hypotenuse

z	1	2	5	10	15	20
tan z	.017	.035	.087	.176	.268	.364

Figure 5 — Tangent table: values of tan z for angles 1° to 20°

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z	1	2	5	10	15	20
tan z	.017	.035	.087	.176	.268	.364
60 tan z	1.02	2.1	5.22	10.56	16.08	21.84

Figure 6 — Expanded table: 60 × tan z compared with z — showing the close numerical correlation

The Tangent Table

Angle z	1°	2°	5°	10°	15°	20°
tan z	0.017	0.035	0.087	0.176	0.268	0.364
60 × tan z	**1.02**	**2.10**	**5.22**	**10.56**	16.08	21.84

Up to about **10°**, the correlation between angle z and 60 × tan z is close: within about 5%, or about half a degree at 10° (60 × tan 10° = 10.56). Most of that 5% comes from using 60 instead of 57.3. Above 15°–20° the error grows, but in practice you should rarely need the rule for angles that large.

> **⚠ Validity Limit**
>
> The 1 in 60 rule is acceptably accurate for angles up to **~20°**. Above this, the tangent relationship becomes significantly non-linear. For track errors larger than 20°, use the nav computer or trigonometry.

5. Expanding or Contracting the Triangle

Fixes don't always occur at exactly 60 NM intervals. The 1 in 60 rule can be adapted for any along-track distance using the principle of **similar triangles**.

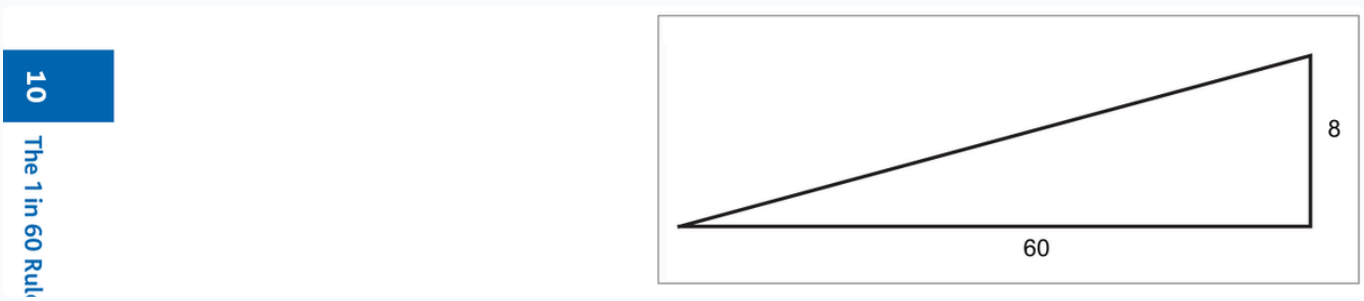


Figure 7 — 8 NM off track in 60 NM — track error = 8°

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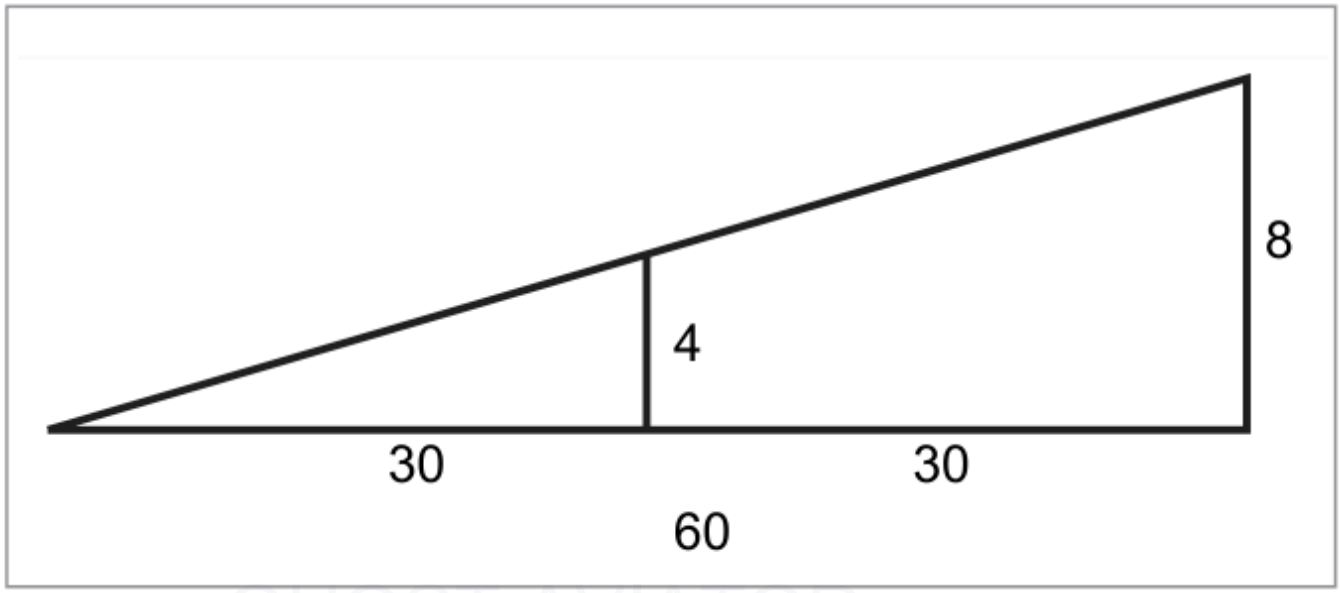


Figure 8 — Same 8° track error at 30 NM along-track gives only 4 NM off track — similar triangles

✎ Scale the Cross-Track Distance

For a given track error angle, the cross-track error scales proportionally with along-track distance:

- 4 NM off in 30 NM = 8 NM off in 60 NM → track error = **8°**
- 10 NM off in 120 NM = 5 NM off in 60 NM → track error = **5°**

6. The Formula & Worked Applications

Track Error Angle

$$z = (\text{Distance Off} / \text{Distance Gone}) \times 60$$

Or rearranged: Distance Off = (z × Distance Gone) / 60

Applications

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Use case	Known	Find	Formula
Track error angle	Distance off, distance gone	Angle z	z = (off/gone) × 60
Cross-track distance	Angle z, distance gone	Distance off	off = (z × gone) / 60
Along-track distance	Angle z, distance off	Distance gone	gone = (off × 60) / z
Glide slope height	GS angle, range	Height	H = (angle × range_ft) / 60
Glide slope range	GS angle, height	Range	range_ft = (H × 60) / angle

✈ TMG vs Track Error

Track Made Good (TMG) = planned track ± track error angle

If aircraft is **left** of track: TMG = planned track − error (track is less than planned)

If aircraft is **right** of track: TMG = planned track + error (track is more than planned)

✎ Glide Slope Worked Example

GS angle 3°, range 2 NM (1 NM = 6 000 ft):

Range = 2 × 6 000 = 12 000 ft

Height = (3 × 12 000) / 60 = **600 ft**

GS angle 2.5°, height 1 000 ft QFE (1 NM = 6 000 ft):

Range (ft) = (1 000 × 60) / 2.5 = 24 000 ft

Range (NM) = 24 000 / 6 000 = **4 NM**

📚 Quick Revision — Chapter 10

- Core rule: adjacent = **60** → opposite (numerically) = angle in degrees
- Formula: **z = (off / gone) × 60**
- Left of track → TMG = planned − z; Right → TMG = planned + z
- Valid up to ~**20°**; 5% error from using 60 instead of 57.3
- Closing angle to destination: same formula using distance from destination
- Glide slope: H(ft) = (GS angle × range_ft) / 60

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Practice Questions & Detailed Answers

12 calculation questions • Track error, TMG, glide slope, closing angle • Full worked solutions

Q1. You are flying from A to B. Position: 60 NM outbound from A, 7 NM left of track. Track error angle?

Anchor: §6 — Formula

> **Answer: 7° Left**

z = (7 / 60) × 60 = **7°** left. At exactly 60 NM, the formula simplifies directly — the off-track distance IS the angle.

> **Instructor's Note:** Source key: 7°L. Verified.

Q2. Flying C to D. Position: 120 NM outbound, 8 NM right of track. Track error angle?

Anchor: §6 — Formula

> **Answer: 4° Right**

z = (8 / 120) × 60 = **4°** right.
At 120 NM, halve the off-track distance to get the angle: 8 ÷ 2 = 4°.

> **Instructor's Note:** Source key: 4°R. Verified.

Q3. Flying E to F. Position: 90 NM outbound, 6 NM right of track. Track error angle?

Anchor: §6 — Formula

> **Answer: 4° Right**

z = (6 / 90) × 60 = 360 / 90 = **4°** right.
At 90 NM, multiply off-track by 2/3: 6 × (60/90) = 4°.

> **Instructor's Note:** Source key: 4°R. Verified.

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Q4. Flying G to H. Position: 30 NM outbound, 4 NM left of track. Track error angle?

Anchor: §6 — Formula

> **Answer: 8° Left**

z = (4 / 30) × 60 = 240 / 30 = **8°** left.
At 30 NM (half of 60), double the off-track: 4 × 2 = 8°.

> **Instructor's Note:** Source key: 8°L. Verified.

Q5. Flying J to K, planned track 045°(T). Position: 80 NM outbound, 4 NM left of track. Track Made Good?

Anchor: §6 — Formula

> **Answer: 042°(T)**

Step 1 — Track error angle: z = (4 / 80) × 60 = 3°
Step 2 — Apply: Aircraft is left of track, so TMG < planned.
TMG = 045° − 3° = **042°(T)**

> **Instructor's Note:** Source key: 042°(T). Verified.

Q6. Flying L to M, planned track 220°(T). Position: 45 NM outbound, 3 NM right of track. Track Made Good?

Anchor: §6 — Formula

> **Answer: 224°(T)**

Step 1 — Track error: z = (3 / 45) × 60 = 4°
Step 2 — Apply: Aircraft is right of track, so TMG > planned.
TMG = 220° + 4° = **224°(T)**

> **Instructor's Note:** Source key: 224°(T). Verified.

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Q7. Flying N to P, planned track 315°(T). Position: 40 NM outbound, 6 NM left of track. Track Made Good?

Anchor: §6 — Formula

> **Answer: 306°(T)**

Step 1 — Track error: z = (6 / 40) × 60 = 9°
Step 2 — Apply: Aircraft is left of track, so TMG < planned.
TMG = 315° − 9° = **306°(T)**

> **Instructor's Note:** Source key: 306°(T). Verified.

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Q8. A surveyor stands 660 metres from a mast and measures an elevation angle of 4° to the top. What is the height of the mast?

Anchor: §6 — Formula

> **Answer: 44 metres**

Distance off: H = (z × distance) / 60 = (4 × 660) / 60 = 2 640 / 60 = **44 m**

The surveyor's 660 m horizontal distance is the 'adjacent' and the mast height is the 'opposite'.

> **Instructor's Note:** Source key: 44 m. Verified.

Q9. Instrument approach, glide slope 3.00°, exactly 2 NM from touchdown. What height should you be passing? (Use 1 NM = 6 000 ft)

Anchor: §6 — Formula

> **Answer: 600 ft**

Step 1 — Convert range to feet: 2 NM × 6 000 = 12 000 ft
Step 2 — Height: H = (3 × 12 000) / 60 = 36 000 / 60 = **600 ft**

> **Instructor's Note:** Source key: 600 ft. Verified.

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Q10. Instrument approach, glide slope 2.5°, correctly on slope, passing 1 000 ft QFE. Range from touchdown? (1 NM = 6 000 ft)

Anchor: §6 — Formula

> **Answer: 4 NM**

Step 1 — Find range in feet: range_ft = (H × 60) / angle = (1 000 × 60) / 2.5 = 60 000 / 2.5 = 24 000 ft

Step 2 — Convert to NM: 24 000 / 6 000 = **4 NM**

> **Instructor's Note:** Source key: 4 NM. Verified.

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Q11. Flying Q to R, planned track 125°(T). Position: 40 NM from R, 2 NM left of track. What track must you fly to arrive overhead R?

Anchor: §6 — Formula

> **Answer: 128°(T)**

Closing angle: z = (2 / 40) × 60 = 3°
Direction: Aircraft is left of track. To close onto R, turn right (increase track).
Track to R = 125° + 3° = **128°(T)**

Note: When flying toward the destination with a closing angle, the correction adds to or subtracts from the planned track to point at R — not at a parallel offset.

> **Instructor's Note:** Source key: 128°(T). Verified.

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Q12. Flying S to T, planned track 272°(T). Position: 50 NM from T, 5 NM right of track. What track must you fly to arrive overhead T?

Anchor: §6 — Formula

> **Answer: 266°(T)**

Closing angle: z = (5 / 50) × 60 = 6°
Direction: Aircraft is right of track. To close onto T, turn left (decrease track).
Track to T = 272° − 6° = **266°(T)**

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> **Instructor's Note:** Source key: 266°(T). Verified.

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The 1 in 60 Formulae

Find	Formula	Notes
Track error angle	**z = (off / gone) × 60**	z in degrees
Cross-track distance	**off = (z × gone) / 60**	Units consistent
Along-track distance	**gone = (off × 60) / z**	—
Glide slope height (ft)	**H = (angle × range_ft) / 60**	1 NM = 6 000 ft for calcs
Glide slope range (ft)	**range_ft = (H × 60) / angle**	Then ÷ 6 000 for NM

Key Facts

Parameter	Value
Exact value (radians)	**57.3°**
Practical approximation	**60°** (~5% error)
Valid angle range	**up to ~20°**
Left of track → TMG	planned – error (TMG less than planned)
Right of track → TMG	planned + error (TMG more than planned)
Closing angle (left)	planned + closing angle
Closing angle (right)	planned – closing angle

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